

G001 Routine Maintenance Procedures

Daily Pre-Use BIT Procedure

Daily Pre-Use BIT Procedure

Prior to commencing any training session, the instructor or maintenance technician must execute the Built-In Test (BIT) sequence to verify system integrity across all simulator subsystems. This procedure ensures that critical components including the visual system, motion platform, flight controls, and avionics simulation are functioning within acceptable parameters.

Begin the BIT sequence with the simulator in a cold and dark state. Ensure all circuit breakers on the instructor station and maintenance panel are set to their normal operating positions. Verify that the motion platform is in the parked position with all six actuators fully retracted and safety pins removed.

BIT Execution Sequence:

1. Power up the instructor operating station (IOS) and wait for the operating system to complete its boot cycle (approximately 90-120 seconds)
2. Access the maintenance menu by entering the technician password at the login prompt
3. Select "System Diagnostics" followed by "Comprehensive BIT"
4. Initiate the automated test sequence and monitor the progress display
5. Review each subsystem as it completes testing: visual system (channels 1-6), motion control system, hydraulic simulation, electrical system simulation, flight control loading system, and avionics suite
6. Document any fault codes in the maintenance logbook using the standardized fault reporting format
7. For any failures indicated by red status indicators, consult the fault isolation manual section corresponding to the affected system
8. Upon successful completion (all green indicators), print the BIT summary report and attach to the daily operations log

The complete BIT cycle typically requires 8-12 minutes. Any subsystem reporting a fault code must be investigated and cleared before releasing the simulator for training operations.

SAMPLE / NOT REAL

Common nuisance faults include IRS alignment timeout (Code 2847) and FMS database version mismatch (Code 3192), both of which typically clear upon system restart.

Weekly Maintenance Checklist

The weekly maintenance checklist ensures all critical simulator systems maintain operational readiness and certification compliance between scheduled quarterly inspections. This routine verification must be completed by qualified maintenance personnel and logged in the facility maintenance tracking system.

Visual System Verification

Begin by powering up the visual system and allowing a minimum 15-minute warm-up period for projector stabilization. Inspect all display surfaces for pixel degradation, color uniformity, and edge blending accuracy. Document any anomalies exceeding $\pm 5\%$ luminance variation. Verify collimated mirror cleanliness and alignment using the optical test pattern generator. Check cooling fan operation on all projector assemblies and confirm exhaust temperatures remain within 45-65°C operating range.

Motion System Inspection

1. Conduct a full motion platform self-test sequence using the maintenance control panel
2. Monitor hydraulic pressure gauges during extension and retraction cycles (normal range: 2800-3000 PSI)
3. Inspect all six actuator boots for tears, leaks, or contamination
4. Verify emergency stop circuits activate within 100ms response time
5. Check accumulator pre-charge pressure (nitrogen: 1850 PSI ± 50)
6. Test motion disable interlocks and instructor override functions

Avionics and Control Loading

Execute the weekly BIT sequence on all flight deck systems including FMC, IRS units, EICAS/ECAM displays, and radio navigation equipment. Control loading systems require torque verification at neutral position—record readings for yoke pitch (± 0.5 lb-ft), yoke roll (± 0.8 lb-ft), and rudder pedals (± 1.2 lb-ft). Any deviation beyond tolerance requires immediate adjustment and recalibration.

Audio System Check

Test all cockpit audio channels including intercom, radio simulation, aural warnings, and environmental sound reproduction. Verify proper operation of the instructor-pilot

communication system and emergency override capability. Sound pressure levels should measure 75-85 dB at pilot headset position during standard callout testing.

Complete the weekly checklist form WM-G001-REV4 and file with the training center quality assurance department within 24 hours of inspection completion.

Parts Ordering and Vendor Coordination

Parts Ordering and Vendor Coordination

All replacement components for the G001 simulator must be sourced through approved vendors listed in the Master Vendor Registry (MVR-G001-Rev4). Prior to initiating any parts order, maintenance personnel shall verify component part numbers against the Illustrated Parts Catalog (IPC) and cross-reference with the current Minimum Equipment List (MEL) to determine operational impact during replacement intervals.

Standard Parts Requisition Procedure:

1. Document the failed component's part number, serial number, and fault code from the BIT diagnostics or EICAS/ECAM fault history
2. Complete Form G001-PR-01 (Parts Requisition) including LRU designation and reason for replacement
3. Verify availability through the primary vendor's online portal or contact technical sales representative
4. Obtain three-digit authorization code from Maintenance Control before submitting orders exceeding \$5,000 USD
5. Submit requisition to Procurement with required lead time notation (standard, expedited, or AOG)
6. Track order status through vendor portal and update maintenance scheduling system accordingly

Critical flight control components including FCC modules, IRS units, and FMS computers require additional coordination with the OEM technical representative. Exchange programs are available for most avionics LRUs, reducing downtime to 24-48 hours for standard shipments. When ordering display units (PFD, ND, EICAS), specify software version compatibility with current simulator baseline configuration documented in the Configuration Management Database (CMDB).

SAMPLE / NOT REAL

For hydraulic actuators, motion system components, and visual system projectors, coordinate delivery schedules with planned maintenance windows to minimize simulator downtime. AOG orders bypass standard procurement channels and require direct contact with the vendor's emergency support line. Document all AOG incidents in the Maintenance Occurrence Report (MOR) system for trend analysis.

Consumable items including hydraulic fluid (MIL-PRF-83282), air filters, and cooling system refrigerant should maintain minimum stock levels per the Consumables Inventory Matrix (CIM-G001). Reorder points are automatically generated when inventory falls below 30-day operational reserves.

Lost Time Categorization Guidelines

Lost Time Categorization Guidelines

Accurate categorization of maintenance-related lost time is essential for regulatory compliance, operational analysis, and resource allocation planning. All maintenance events resulting in simulator unavailability must be documented in the Maintenance Action Tracking System (MATs) using the appropriate lost time category codes.

Category Definitions and Application

Category 1 - Scheduled Maintenance (SM): Planned downtime for routine inspections, preventive maintenance, or software updates. This includes quarterly BIT verification runs, annual IOS recalibration, and scheduled visual system lamp replacements. SM events must be coordinated with training scheduling at least 72 hours in advance.

Category 2 - Unscheduled Corrective Maintenance (UCM): Unexpected failures requiring immediate attention that render the simulator non-operational or training-limited. Examples include FMS database corruption, motion system hydraulic leaks, IOS processor faults, or EICAS display failures. UCM time begins when the fault is first reported and ends when the system is returned to full operational status.

Category 3 - Modification/Upgrade (MU): Time allocated for installation of hardware modifications, software version upgrades, or configuration changes. This includes cockpit equipment retrofits, visual database updates, or flight model refinements. MU activities require advance approval from the Training Standards Office.

SAMPLE / NOT REAL

Category 4 - External Delay (ED): Downtime caused by factors beyond maintenance control, such as facility power outages, HVAC system failures affecting simulator environmental requirements, or vendor support delays exceeding 24 hours.

Documentation Procedure

1. Initiate MATS work order immediately upon fault detection or scheduled maintenance commencement
2. Select appropriate category code from dropdown menu
3. Record start time using facility UTC reference clock
4. Document all troubleshooting steps, fault codes retrieved, and corrective actions performed
5. Verify system functionality through appropriate BIT sequences before closing work order
6. Record completion time and calculate total lost time duration
7. Obtain supervisor approval for all UCM events exceeding 4 hours

All Category 2 events must include root cause analysis documentation within 48 hours of resolution.